



Zyro

PAKISTAN'S FIRST AI COMMERCE OS

TECHNICAL RESEARCH & DEVELOPMENT REPORT

Platform & Courier API Integration Analysis

A comprehensive technical evaluation of seven commerce API integrations — covering authentication methods, endpoint capabilities, pricing models, known limitations, and integration recommendations for the Zyro Commerce OS platform.



Daraz



Shopify



TCS Courier



Leopards



M&P



Pakistan Post



DHL Express

DATE

July 2026

VERSION

1.0

INTEGRATIONS

7 Platforms

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CONFIDENTIAL

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PRICING LEGEND USED IN THIS REPORT

PAID

Requires a paid subscription or account fee

FREE

API access is free (account registration required)

PER USE

No subscription; charged per shipment / per API call

01

Executive Summary

High-level overview of all seven integrations

This report documents a technical research and development study conducted by the Zyro Engineering team, evaluating seven commerce platform and courier APIs for integration into the Zyro AI Commerce OS. Each integration is assessed across authentication mechanism, API endpoint coverage, pricing model, data availability, known limitations, and implementation complexity.

7

Integrations
Analysed

2

Marketplace
Platforms

5

Courier Providers

3

Auth Mechanisms

2

Webhook-Capable
APIs

Pricing Summary at a Glance

Integration	API Access Cost	Account Type	Per-Use Charges
Daraz	Free	Approved Daraz Seller Account	None (marketplace commission applies)
Shopify	PAID — from \$29/month USD	Shopify Store Subscription Required	Transaction fees on some plans
TCS Courier	Free	TCS Merchant/Corporate Account	Per-shipment charges (PKR ~250–500)
Leopards	Free	Leopards Merchant Account	Per-shipment charges (PKR ~200–400)
M&P	Free	M&P Merchant Account (volume req.)	Per-shipment charges (PKR ~200–450)
Pakistan Post	No API — TrackingMore: \$9+ /month	TrackingMore Developer Account	Per-tracking after free tier (100/month)
DHL Express	Free (MyDHL API)	DHL Corporate Account Required	Per-shipment (PKR ~1,500–3,000+)

⚠ CRITICAL FINDING — DARAZ DATAMOAT

Daraz masks customer names and phone numbers for all sellers by default. A **Sensitive Data Privilege (SDP)** application is required to access real customer PII. Approval takes 2–4 weeks and is not guaranteed. Without SDP, any CRM or customer communication features built on Daraz data will not function.

✓ RECOMMENDED LAUNCH PRIORITY

Shopify → TCS → Leopards → Daraz → DHL → M&P → Pakistan Post. Shopify has the cleanest API with full real-time webhooks. TCS and Leopards cover 90%+ of Pakistan e-commerce domestic shipments. Daraz should follow after the DataMoat SDP approval.

E-Commerce Platforms

Daraz Open Platform · Shopify Admin API



2.1 — Daraz Open Platform

Marketplace API · Pakistan, Bangladesh, Sri Lanka, Nepal, Myanmar

FREE
API

Seller Account
Required

HMAC-
SHA256

ACCOUNT & API PRICING

API Access: Free — No monthly fee to use the Daraz Open Platform API.

Account Requirement: You must be a registered and approved Daraz seller. Seller registration is free at seller.daraz.pk but requires business documentation and approval (1–5 business days).

Sensitive Data Privilege (SDP): Additional application needed to access real customer names and phone numbers. No fee, but approval is not guaranteed and takes 2–4 weeks.

Revenue Model: Daraz charges sellers a commission (8–15%) per sale. No API call charges.

DEVELOPER PORTAL

open.daraz.com

PK BASE URL

<https://api.daraz.pk/rest>

AUTH METHOD

HMAC-SHA256 Request
Signing

WEBHOOK SUPPORT

None — Pull Only

MAX RECORDS/REQUEST

100 orders (offset pagination)

ZYRO PLATFORM ID

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REQUIRED CREDENTIALS (STORED IN SUPABASE USERS TABLE)

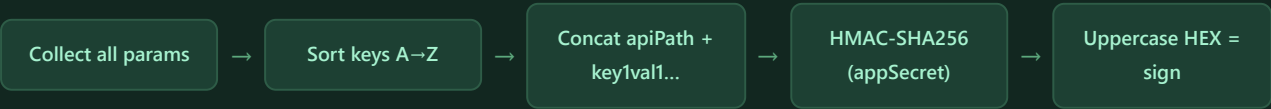
- daraz_app_key

→ Numeric App Key from App Console (e.g. 123456)
- daraz_app_secret

→ 32-character hex secret — used to compute HMAC-SHA256 signatures
- daraz_access_token

→ Long alphanumeric token from Self-Authorization flow — may expire

HMAC-SHA256 Signing Flow (Every Request)



KEY API ENDPOINTS

- GET

/orders/get

List orders — status, date range, offset, limit (max 100)

GET`/orders/items/get`

Fetch line items for an order (separate call required)

GET`/order/get`

Single order by order_id

GET`/products/get`

Product catalogue with SKUs and inventory

POST`/orders/packing/set`

Mark order as packed — triggers Daraz fulfilment workflow

GET`/finance/transaction/details/get`

Settlement and payment transaction records

✓ STRENGTHS

- ✓ Dominant marketplace in Pakistan — very high order volume
- ✓ Covers 5 South Asian countries under one API
- ✓ Product catalogue, inventory and finance APIs included
- ✓ API access is completely free for sellers

✗ LIMITATIONS

- ✗ No webhook — requires polling or cron jobs
- ✗ Customer PII masked without SDP approval
- ✗ Order items need a second separate API call
- ✗ Search limited to Order ID only



2.2 — Shopify Admin API

E-Commerce Platform API · Global (195+ countries)

PAID PLATFORM

FREE API

Webhooks ✓

ACCOUNT & API PRICING — PAID

Shopify Store Subscription is **REQUIRED** and **PAID**:

- **Basic Plan:** \$29 USD/month — 2 staff accounts, basic reports
- **Shopify Plan:** \$79 USD/month — 5 staff accounts, professional reports
- **Advanced Plan:** \$299 USD/month — 15 staff accounts, advanced reports
- **Plus Plan:** From \$2,300 USD/month — enterprise

Admin API Access: FREE — Once a store subscription exists, creating a Custom App and using the Admin API incurs no additional charge. There are no per-API-call fees.

Transaction Fees: 2% on Basic/Shopify plans if not using Shopify Payments; 0.5% on Advanced.

API VERSION

2024-01 (REST + GraphQL)

BASE URL

`{store}.myshopify.com/admin/api/2024-01`

AUTH METHOD

X-Shopify-Access-Token header

WEBHOOK SUPPORT

Yes — HMAC-SHA256 verified

MAX RECORDS/REQUEST

250 orders (cursor pagination)

ZYRO PLATFORM ID

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REQUIRED CREDENTIALS

shopify_store_domain

→ mystore.myshopify.com — without https://

shopify_access_token

→ Admin API token beginning with **shpat_** — shown once on app install

shopify_webhook_secret

→ Optional — signing secret for verifying incoming webhook HMAC signatures

Custom App Setup (One-Time, Free)

Admin → Settings
→ Apps

Develop Apps →
Create

Set API Scopes
(read_orders,
read_customers)

Install App

Copy shpat_token

KEY API ENDPOINTS

GET

/orders.json

List orders with line items, customer, billing — max 250 per request

GET`/orders/count.json`

Total order count — fast endpoint for stats, avoids fetching full orders

GET`/customers.json`

Full customer list with PII — name, phone, email, address (never masked)

POST`/webhooks.json`

Register webhook. Topics: orders/create, orders/updated, orders/paid, orders/cancelled

POST`/fulfillments.json`

Mark order as shipped with tracking number and courier name

GET`/products.json`

Product catalogue with variants, SKUs, inventory levels

i RATE LIMITS — LEAKY BUCKET MODEL

Bucket size: **40 requests**, refill at **2 requests/second**. Every response includes `X-Shopify-Shop-Api-Call-Limit: used/max`. On HTTP 429, wait for the `Retry-After` header value before retrying.

✓ STRENGTHS

- ✓ Real-time webhooks — orders arrive in under 2 seconds
- ✓ Full customer PII always available (no masking)
- ✓ Line items included in the same order response
- ✓ Simple header-based auth — no signing algorithm
- ✓ 250 orders per request (2.5× Daraz's limit)
- ✓ Excellent official documentation and SDKs

✗ LIMITATIONS

- ✗ Store subscription is PAID (\$29–\$299/month USD)
- ✗ Access token shown only once — must be saved immediately
- ✗ Revenue may be in non-PKR currencies
- ✗ Webhook secret management adds operational work

03

Courier Services

TCS · Leopards · M&P · Pakistan Post · DHL Express



3.1 — TCS Courier

Pakistan's Largest Courier · Est. 1983 · 300+ Cities

FREE API

PER SHIPMENT

Bearer Token

ACCOUNT & API PRICING

API Access: Free — TCS does not charge for API integration access.

Account Requirement: Must register as a TCS Corporate/Merchant client. Contact TCS merchant services (tcscourier.com) with your business details. Registration is free but requires a formal application.

Per-Shipment Rates (approximate): PKR 250–500 for standard parcels within Pakistan. COD collection service is available — TCS remits collected cash within 5–7 business days.

Minimum Volume: Some corporate accounts require a minimum monthly shipment commitment.

DEV URL

devconnect.tcscourier.com

PROD URL

ociconnect.tcscourier.com

AUTH

Bearer Token (2-step login)

ORDER LIST API

Not Available

COD SUPPORT

Yes

ZYRO PLATFORM ID

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REQUIRED CREDENTIALS

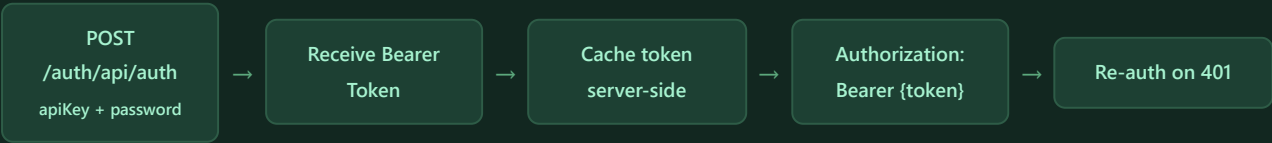
- tcs_api_key

→ TCS-issued username / API Key
- tcs_api_password

→ Password paired with the API key
- tcs_cost_centre_code

→ Merchant Cost Centre Code — required on every shipment booking request

Two-Step Authentication Flow



KEY API ENDPOINTS

POST

/auth/api/auth

Login — returns Bearer token. Required before any other call

POST`/cod/api/v1/shipments`

Book shipment — returns CN number (Consignment Note)

GET`/cod/api/v1/tracking/{cn}`

Track shipment by CN — full event timeline

POST`/cod/api/v1/shipments/cancel`

Cancel a booking before pickup

GET`/cod/api/v1/cities`

List all 300+ serviceable cities with codes

⚠ NO ORDER LIST API

TCS provides no endpoint to retrieve all booked shipments. Order history and statistics must be maintained entirely in your own Supabase database (filter by `platform_id = 3`). The TCS API is booking-and-tracking only.



3.2 — Leopards Courier

Domestic Courier · Pakistan · 400+ Cities · COD + Express

FREE API

PER SHIPMENT

Test Mode

ACCOUNT & API PRICING

API Access: Free — No charge for API integration. Access is granted upon merchant account approval.

Account Requirement: Register at leopardscourier.com merchant portal. Free registration; requires valid CNIC, business address, and bank account for COD remittances.

Per-Shipment Rates (approximate): PKR 200–400 for standard parcels. COD remittance within 5 business days.

Test Mode: Built into every API call via `enable_test_mode: "1"` — no separate sandbox account needed.

BASE URL

`merchantapi.leopardscourier.com`

AUTH METHOD

API Key + Password in request body

TEST MODE

`enable_test_mode: "1"`

ORDER LIST API

Limited (recent only)

COD SUPPORT

Yes

ZYRO PLATFORM ID

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REQUIRED CREDENTIALS

`leopards_api_key`

→ API Key from Leopards merchant portal

`leopards_api_password`

→ API Password — sent in every request body (no separate auth step needed)

KEY API ENDPOINTS

POST

`/api/v1/authenticate_user`

Verify credentials on initial connection

POST

`/api/v1/add_packet`

Book shipment — returns packet tracking number

POST

`/api/v1/track_packet`

Track by tracking number — full event history

POST

`/api/v1/cancel_packet`

Cancel booking before pickup

POST

`/api/v1/get_cities`

All 400+ serviceable cities with codes — required for `add_packet`

POST

`/api/v1/get_packet_list`

Recent packet bookings list (limited date window)

✓ **STRENGTHS**

- ✓ 400+ cities — widest domestic courier coverage
- ✓ Built-in test mode in every call — easy development
- ✓ No separate auth step — simpler integration
- ✓ Cities API for destination code lookup
- ✓ COD and express shipment options

✗ **LIMITATIONS**

- ✗ Credentials in request body (not headers) — slightly less secure
- ✗ No complete order history API
- ✗ No webhook or push notification support
- ✗ Status codes returned in JSON body, not HTTP status



3.3 — M&P Courier Services

Domestic Courier · Pakistan · Same-Day & Next-Day Delivery

FREE API

PER SHIPMENT

Volume Req.

ACCOUNT & API PRICING

API Access: Free — No charge for API access after merchant onboarding.

Account Requirement: M&P's merchant API is not self-serve. You must contact M&P merchant services directly via mnpcourier.pk or their helpline. A minimum monthly shipment volume commitment is typically required.

Per-Shipment Rates (approximate): PKR 200–450 for standard parcels. Same-day delivery available in Karachi, Lahore, and Islamabad at premium rates.

COD: Available. Remittance within 5–7 business days.

Documentation: Provided only after merchant onboarding — not publicly available online.

MERCHANT PORTAL

mnpcourier.pk

AUTH METHOD

API Key in header or body

WEBHOOK SUPPORT

None

SAME-DAY DELIVERY

Yes (major cities)

COD SUPPORT

Yes

ZYRO PLATFORM ID

7 (suggested)

REQUIRED CREDENTIALS

mp_api_key

→ API Key issued by M&P during merchant onboarding

mp_account_number

→ M&P merchant account number — required in all booking requests

KEY API ENDPOINTS

POST

/api/bookshipment

Book shipment — returns CN tracking number

GET

/api/trackshipment/{cn}

Track shipment by CN — status and event history

POST

/api/cancelshipment

Cancel before pickup

GET

/api/getcities

List serviceable cities

POST

/api/getlabel

Generate shipping label PDF for a booked CN

⚠ API ACCESS — NOT SELF-SERVE

Unlike TCS and Leopards where merchant accounts can be registered online, M&P's API access requires a direct business relationship and negotiation with their merchant team. Plan for a 1–2 week onboarding lead time. Request API documentation during the onboarding call.



3.4 — Pakistan Post

Government Postal Service · Nationwide Coverage incl. Remote Areas

TRACKING
PAID

No Booking
API

Via
TrackingMore

ACCOUNT & API PRICING

Pakistan Post API: Does Not Exist — Pakistan Post has no public merchant booking API. Shipments must be dropped at post office counters or via corporate bulk mail agreements.

Tracking via TrackingMore API (Recommended — PAID after free tier):

- **Free Tier:** 100 trackings/month — suitable for initial testing
- **Starter Plan:** \$9 USD/month — 1,000 trackings/month
- **Growth Plan:** \$29 USD/month — 5,000 trackings/month
- **Professional Plan:** \$99 USD/month — 20,000 trackings/month

TrackingMore covers 1,200+ carriers (TCS, Leopards, DHL, Pakistan Post, etc.) under one API key — good value as a unified tracking solution.

TRACKING PORTAL

track.pakpost.gov.pk

TRACKINGMORE API

api.trackingmore.com

CARRIER CODE

pakistan-post

TRACKING FORMAT

EE123456789PK (13-char EMS)

BOOKING API

None Available

COD SUPPORT

Not Available

TRACKINGMORE ENDPOINTS FOR PAKISTAN POST

POST

/v4/trackings/create

Register tracking. Body: { tracking_number, courier_code: "pakistan-post" }

GET

/v4/trackings/{courier_code}/{number} Get tracking events for a specific parcel

POST

/v4/trackings/batch

Batch track up to 40 parcels in one request



3.5 — DHL Express (MyDHL API)

International Express Courier · 220+ Countries · Pakistan → World

FREE
API

PER
SHIPMENT

HTTP Basic
Auth

🔑 ACCOUNT & API PRICING

MyDHL API Access: Free — No subscription or API fee. Access requires a DHL corporate account.

Account Requirement: Apply for a DHL corporate account at developer.dhl.com. A DHL account number (8-digit) is issued after business verification (2–5 business days).

Per-Shipment Rates: DHL Express is a premium international courier. Rates from Pakistan are approximately **PKR 1,500–3,000+** for a 0.5 kg parcel. Rates vary by destination, weight, and dimensions. Use the `/rates` endpoint to get live quotes.

Sandbox: Full sandbox environment available at no cost — test endpoint:
<https://express.api.dhl.com/mydhlapitest>

API NAME

MyDHL API

BASE URL

`express.api.dhl.com/mydhlapitest`

AUTH METHOD

HTTP Basic Auth
(apiKey:apiSecret)

RATE QUOTE API

Yes — `/rates` endpoint

COD SUPPORT

No

ZYRO PLATFORM ID

8 (suggested)

REQUIRED CREDENTIALS

dhl_api_key	→ DHL MyDHL API username (from developer.dhl.com)
dhl_api_secret	→ DHL API password — used in Base64 Basic Auth header
dhl_account_number	→ 8-digit DHL corporate account number — required for booking and rate quotes

KEY API ENDPOINTS

POST	<code>/shipments</code>	Create shipment — returns AWB number + label PDF URL
GET	<code>/tracking?shipmentTrackingNumber=</code>	Real-time tracking with granular event data
GET	<code>/rates</code>	Live rate quotes — show customers exact shipping cost before booking
POST	<code>/pickups</code>	Schedule DHL pickup from your address
GET	<code>/address-validate</code>	Validate delivery address serviceability before booking

04

Comparison Matrix

All seven integrations evaluated across key technical and commercial dimensions

E-Commerce Platforms

Platform	API Cost	Webhooks	Customer PII	Max per Request	Search	Auth
Daraz	Free API	None	Masked (SDP needed)	100 orders	Order ID only	HMAC-SHA256
Shopify	\$29+/month store	Full support	Always visible	250 orders	Name / email / order#	Token header
WooCommerce	Free (plugin)	Push webhook	Always visible	100 orders	Name / email / phone	Basic Auth

Courier Services

Courier	API Cost	Acct. Type	Book API	Track API	COD	Coverage	Est. Rate
TCS	Free	Corporate acct.	Yes	Yes	Yes	300+ cities	PKR 250–500
Leopards	Free	Merchant acct.	Yes	Yes	Yes	400+ cities	PKR 200–400
M&P	Free	Volume commit.	Yes	Yes	Yes	Major cities	PKR 200–450
Pakistan Post	No API / \$9+/mo tracking	TrackingMore	None	Via 3rd party	No	All Pakistan	PKR 80–250
DHL Express	Free API	Corporate acct.	Yes	Yes	No	220+ countries	PKR 1,500+

Implementation Effort & Recommended Priority

Integration	Est. Dev Time	API Quality	Documentation	Business Value	Priority
Shopify	2–3 days	Excellent	Excellent	Very High	P1 — Now
TCS	3–4 days	Good	Moderate	Very High	P1 — Now
Leopards	2–3 days	Good	Moderate	High	P2 — Soon

INTEGRATION	EST. DEV TIME	API QUALITY	DOCUMENTATION	BUSINESS VALUE	PRIORITY
Daraz	4–6 days	Moderate	Moderate	Very High	P2 — After SDP
DHL Express	3–4 days	Excellent	Excellent	Medium	P3 — Later
M&P	2–3 days + negotiation	Moderate	Limited	Medium	P3 — Later
Pakistan Post	1 day (tracking only)	Very Limited	None	Low	P4 — Optional

05

Integration Recommendations

Actionable guidance for the Zyro engineering and product teams

1 Shopify — Use Webhooks, Not Polling

Shopify's `orders/create webhook` delivers new orders in under 2 seconds. Register the webhook programmatically via the `POST /webhooks.json` endpoint on first connection and store the webhook ID per user for deregistration. Verify every incoming webhook with HMAC-SHA256 using the stored webhook secret to prevent spoofed requests. This removes the need for any polling loop.

2 TCS — Cache the Bearer Token Server-Side

TCS requires a two-step login to obtain a Bearer token before every session. Cache this token in a server-side in-memory object keyed by user ID. On a 401 response from any TCS endpoint, automatically re-authenticate and retry the original request once. This prevents unnecessary login calls on every API request and reduces latency for end users.

3 Daraz — Apply for Sensitive Data Privilege Immediately

The Daraz DataMoat system makes customer CRM features non-functional without the Sensitive Data Privilege (SDP). Apply on the Daraz App Console as soon as the integration is live — approval takes 2–4 weeks. In the meantime, display a clear "Customer data protected — SDP approval pending" message in the Daraz order view so users understand the limitation without confusion.

4 All Pakistan Couriers — Standardise Phone Normalisation

TCS, Leopards, and M&P all require phone numbers in `03xxxxxxxxx` format — 10 digits, no country code, no +92 prefix. Implement a single shared `normalizePhone()` utility that strips spaces, dashes, +92, and 0092 prefixes before any courier booking call. Unnormalised phone numbers cause silent booking failures where the API accepts the request but the delivery attempt fails.

5 Pakistan Post — Integrate TrackingMore, Not a Custom Solution

Pakistan Post has no merchant API. Rather than a custom web scraping solution (fragile, maintenance-heavy), integrate **TrackingMore API**. A single TrackingMore key also covers TCS, Leopards, DHL, and 1,200+ other carriers — making it a universal tracking fallback for any parcel entered manually by a user. Start on the free tier (100/month) and upgrade as usage grows.

6 DHL — Position Explicitly for International Shipments

DHL rates (PKR 1,500–3,000+) are 5–10× higher than domestic couriers. Surface DHL automatically when a customer's delivery address is outside Pakistan — filter by country code in the order normaliser. Use the `GET /rates` API to show the live shipping cost to the user before they confirm the DHL booking, which prevents billing surprises and reduces cancellations.

7 Daraz & Pull-Only Platforms — Implement Vercel Cron Sync

For Daraz (no webhooks), implement a cron job at `/api/cron/daraz` running every 30 minutes. Query Supabase for all users with Daraz credentials configured, then pull and upsert their orders. Rate-limit the cron to 5 users/minute to respect Daraz's ~60 requests/minute threshold. Add a `last_synced_at` column per user to track freshness and show users when their data was last updated.

06

Supabase Column Reference

All credential columns to be added to the users table via ALTER TABLE migrations

MIGRATION PATTERN

Run: ALTER TABLE users ADD COLUMN IF NOT EXISTS {column_name} TEXT; — the IF NOT EXISTS guard makes these migrations safe to re-run.

COLUMN NAME	PLATFORM	DESCRIPTION	REQUIRED?
<code>daraz_app_key</code>	Daraz	Numeric App Key from Daraz Open Platform App Console	Yes
<code>daraz_app_secret</code>	Daraz	32-char hex secret used to compute HMAC-SHA256 signatures	Yes
<code>daraz_access_token</code>	Daraz	Long alphanumeric token from Self-Authorization flow	Yes
<code>shopify_store_domain</code>	Shopify	mystore.myshopify.com — without https://	Yes
<code>shopify_access_token</code>	Shopify	Admin API token starting with shpat_ — shown once on install	Yes
<code>shopify_webhook_secret</code>	Shopify	Webhook signing secret — for HMAC-SHA256 verification of incoming events	Recommended
<code>tcs_api_key</code>	TCS	TCS username / API Key	Yes
<code>tcs_api_password</code>	TCS	TCS API password (used in Bearer token login)	Yes
<code>tcs_cost_centre_code</code>	TCS	Merchant Cost Centre Code — required on every TCS booking request	Yes
<code>leopards_api_key</code>	Leopards	Leopards merchant API key	Yes
<code>leopards_api_password</code>	Leopards	Leopards API password — sent in every request body	Yes
<code>mp_api_key</code>	M&P	M&P API key issued during merchant onboarding	Yes
<code>mp_account_number</code>	M&P	M&P merchant account number required in all bookings	Yes
<code>dhl_api_key</code>	DHL	DHL MyDHL API username from developer.dhl.com	Yes
<code>dhl_api_secret</code>	DHL	DHL API password for HTTP Basic Auth header	Yes
<code>dhl_account_number</code>	DHL	8-digit DHL corporate account number	Yes
<code>trackingmore_api_key</code>	Pakistan Post	TrackingMore API key — also usable as universal tracking fallback	Yes

